Question			Answer	Marks	AO	Guidance
11	(a)		$\dfrac{\sqrt{n+1}+1-(\sqrt{n+1}-1)}{(\sqrt{n+1}-1)(\sqrt{n+1}+1)}$	M1	3.1a	• Finding common denominator or attempt to rationalise denominator for at least one fraction e.g. $\frac{1}{\sqrt{n+1}-1}\cdot\frac{\sqrt{n+1}+1}{\sqrt{n+1}+1}-\frac{1}{\sqrt{n+1}+1}\cdot\frac{\sqrt{n+1}-1}{\sqrt{n+1}-1}$ *or* $\frac{\sqrt{n+1}+1}{n}-\frac{(\sqrt{n+1}-1)}{n}$ oe • Condone consistently considering any general term in the sequence e.g. a_{k+1} for all marks $\dfrac{1}{\sqrt{k+2}-1}\cdot\dfrac{\sqrt{k+2}+1}{\sqrt{k+2}+1}-\dfrac{1}{\sqrt{k+2}+1}\cdot\dfrac{\sqrt{k+2}-1}{\sqrt{k+2}-1}$
			$\dfrac{2}{(n+1)-1}=\dfrac{2}{n}$	A1	2.1	• Correct simplification • a_{k+1} leads to $\frac{2}{k+1}$
			Numerator and denominator are integers (and denominator is not zero) so this is rational	A1	2.2a	• nfww • Correct explanation of why the general term must be rational e.g. in form $\frac{p}{q}$ oe p, q integers • Condone rational as n an integer or k integer • '$\frac{2}{n}$ is rational' alone is **A0** - <u>must</u> include comment that n is an integer
				[3]		

Question			Answer	Marks	AO	Guidance
11	(b)		(Since $n > 0$) as n increases, $\frac{2}{n}$ decreases Or $\frac{2}{n+1} - \frac{2}{n} = -\frac{2}{n(n+1)}$	**M1**	**2.2a**	• Must relate to $\frac{2}{n}$, $a_n \to 0$ is not sufficient • Allow "($n > 0$ and) $\lim_{n\to\infty} \frac{2}{n} = 0$" • Condone omission of $n > 0$ • May find the difference between 2 terms or use inequalities • Allow $f(x) = \frac{2}{x} \Rightarrow f'(x) = -\frac{2}{x^2} < 0$ (condone n instead of x) • Condone listing terms for **M1 A0** • **FT** *their* $\frac{2}{n}$ provided in the form $\frac{b}{n+c}$ where b, c integers
			⇒ sequence is decreasing	**A1**	**2.5**	**See appendix** • **Convincing** conclusion e.g. difference between terms is negative, so sequence is decreasing or $\frac{2}{n+1} < \frac{2}{n}$, so decreasing or function $f(x)$ is decreasing therefore sequence a_n decreasing • **A1** is only available if *their* $\frac{2}{n}$ is correct for *their* term e.g. $\frac{2}{k+1}$ from considering a_{k+1} • **SC B1** for 'decreasing' following **M0** for a correct but incomplete argument. But not just for 'decreasing' unsupported.
				[2]		

Question 11b

These would earn M1A1
As n tends towards infinity $\frac{2}{n}$ will tend towards zero. Decreasing.
$a_{n+1} = \frac{2}{n+1} < \frac{2}{n} = a_n$ So $a_{n+1} < a_n$ so the sequence is decreasing
As k increases $\frac{2}{k+1}$ decreasing, therefore the sequence is decreasing **Note: This is follow through from their answer in 11a as noted in the guidance column.**

This would earn SCB1
$n = 1 \rightarrow a_1 = 2$ $n = 2 \rightarrow a_2 = 1$ $n = 2 \rightarrow a_3 = \frac{2}{3}$ $n = 4 \rightarrow a_4 = \frac{1}{2}$ It is decreasing. **This is insufficient for the award of M1A1 marks as they are using numerical values, however would earn SCB1.**

This would earn M0A0
The sequence is decreasing **No special case awarded as 'decreasing' is unsupported.**